College Board AP® Chemistry

Your notes

Lewis Diagrams

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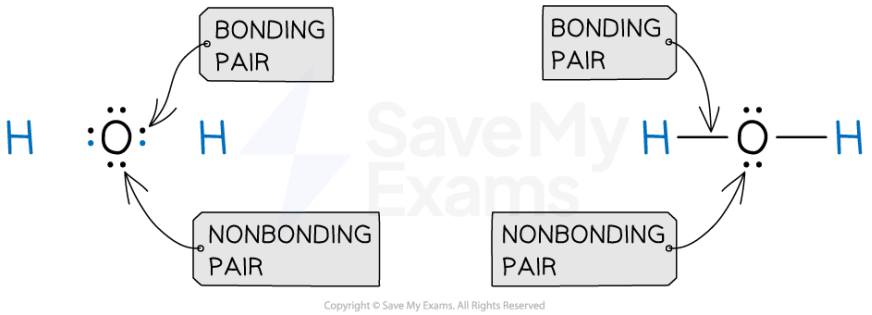
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Lewis Diagrams

- A **Lewis Diagram** is used to represent molecular compounds by showing the valence electrons around atoms
- Electrons are drawn in pairs and they are represented by For example, the water molecule can be represented as:

Lewis Diagrams for Water



Different Lewis Diagrams for the water molecule showing bonding pairs as lines or dots

- Lewis diagrams must show all the electron pairs even if they are **bonding** or **nonbonding** electrons
  - A **bonding pair** is a shared pair of electrons
  - A **nonbonding pair** is a lone pair of electrons that is not involved in bonding
- **Double bonds** are represented with two lines connecting the two atoms involved (=)
  - This implies that 4 electrons are being shared between them
- **Triple bonds** are represented with three lines connecting the two atoms involved (≡)
  - This implies that 6 electrons are being shared between them
- Atoms form covalent compounds to have a full valence shell of electrons
- The **octet rule** is the tendency that explain why atoms are most stable when they complete their valence shell with **8 electrons**
- However, there are some exceptions to the octet rule:
  - The hydrogen atom (H) completes a This means it can only complete its valence shell with **2 electrons**
  - The boron atom (B) completes a This means it can only complete its valence shell with **6 electrons**

Steps for drawing Lewis Diagrams

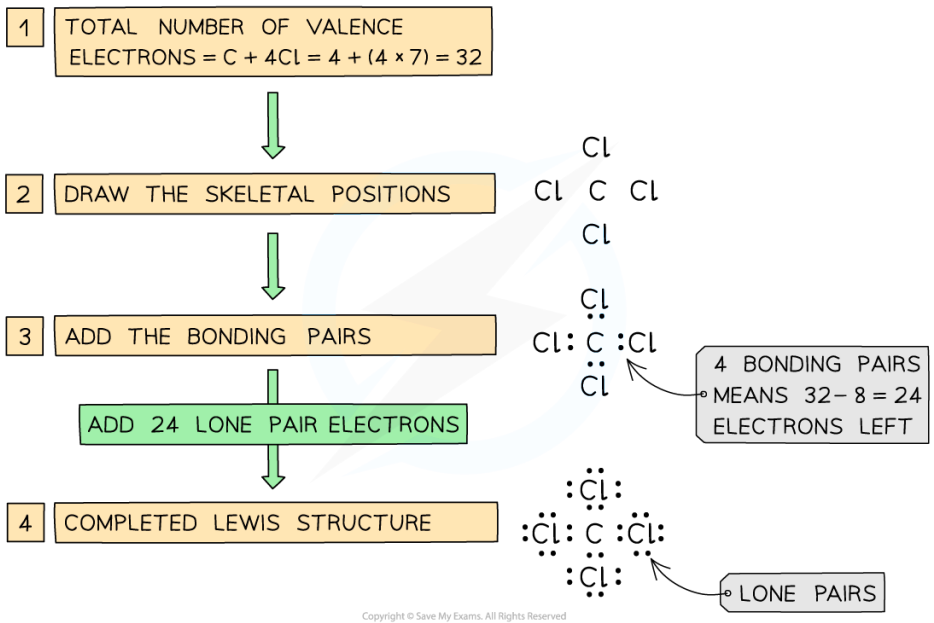
1. Calculate the **total number of valence electrons** using your periodic table
2. Draw a correct skeletal structure to show how the atoms are linked between them
3. Distribute one pair of electrons between each atom to represent single bonds
4. Add electron lone pairs to complete octets (except H which completes two valence electrons). It is a good practice to start on terminal atoms and finish with the central atom
5. If there are still atoms which lack an octet, form double or triple bonds
6. When the structure is finished, check if all the atoms have completed an **octet** (except from H) and check if the amount of electrons distributed is the same as the **total number of valence electrons**

Your notes

Worked Example

Draw a Lewis structure for CCl_4

Answer:



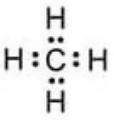
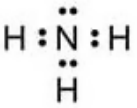
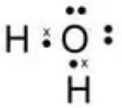
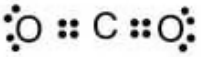
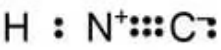
Steps in drawing the Lewis Structure for CCl_4

Further examples of drawing Lewis Diagrams

- By following six simple steps, the Lewis Diagrams of these common molecules can be drawn

Molecule	Total number of valence electrons	Lewis Diagram

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CH_4	C + 4H 4 + (4 x 1) = 8	
NH_3	N + 3H 5 +(3 x 1)=8	
H_2O	2H + O (2 x 1) + 6 =8	
CO_2	C + 2O 4 + (2 x 6) = 16	
HCN	H+C+N 1+ 4 + 5 = 10	

Lewis diagrams for common molecules

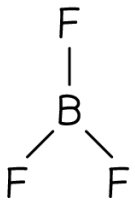
Worked Example

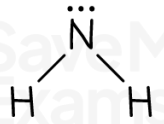
Which of the following Lewis electron-dot diagrams is drawn incorrectly?

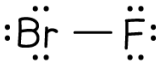
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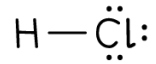
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A 

B 

C 

D 

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Answer:

- The correct answer is **B.**
- **This is** because the nitrogen atom does not complete its octet. It has just 6 electrons, 2 bonding pairs and 1 nonbonding pair. Therefore, the Lewis Diagram is not correctly drawn.
- The boron atom in **A** completes its **sextet**. The fluorine atoms in **A** and **C,** and the bromine atom in **C,** and the chlorine atom in **D** complete their **octets.** In the case of **D,** all hydrogen atoms complete their **duet.** Therefore, their Lewis Diagrams are drawn correctly

Your notes

Resonance & Lewis Structures

- Some molecules can be represented by more than one Lewis structure
- **Resonance structures** are different variations of **the same molecule**, but with **different arrangement of electrons**
- The **resonance hybrid** is the actual structure of a molecule, and it is a mixture of all its resonance structures
- Resonance occurs because electrons are often **delocalized**
- Delocalized electrons are not fixed to a single atom or covalent bond, therefore they can be spread through the molecule
- The **resonance hybrid** is the most stable form of a molecule because the delocalization of electrons stabilize the structure

Resonance structures of the nitrate(V) ion

- To draw the resonance structure of the nitrate(V) ion, the first step is to count the valence electrons that are present in the ion
- There is one nitrogen atom, three oxygen atoms and a negative charge
  - Nitrogen in group 15 contributes with 5 valence electrons
  - Oxygen in group 16 contributes with 6 valence electrons
  - The negative charge means that an electron has been gained

$$\text{Valence electrons} = \text{N} + (3 \times \text{O}) + 1\,\text{e}^-$$

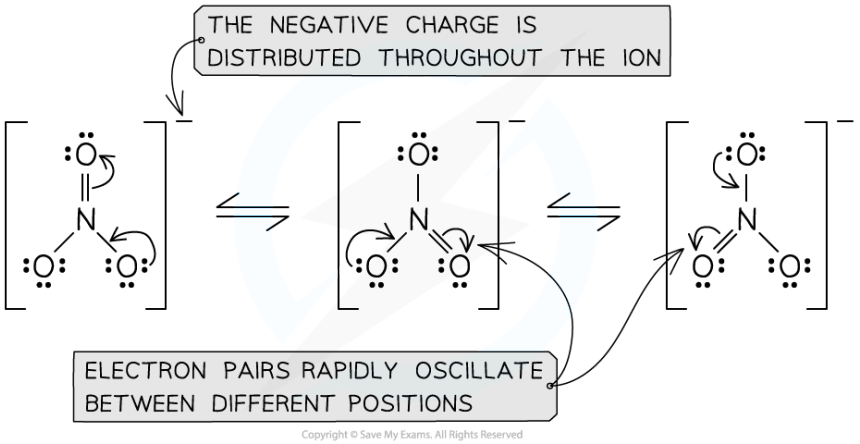
$$\text{Valence electrons} = 5 + (3 \times 6) + 1\,\text{e}^-$$

$$\text{Valence electrons} = 24\ \text{e}^-$$

- After drawing an skeletal structure (N as the central atom) and placing the bonding and nonbonding electrons, three Lewis structures are possible
- These Lewis structures consist of a double bond and two single bonds. However, a pair of electrons from the double bond and lone pairs of electrons can oscillate between three different positions
- These oscillation allows the double bond to migrate from one part of the molecule to another

Resonance in the Nitrate(V) ion

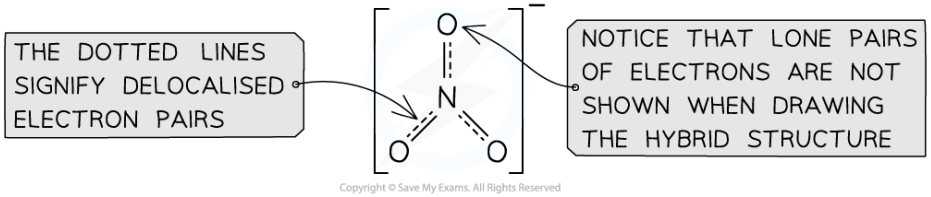
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Resonance structures in the nitrate ion

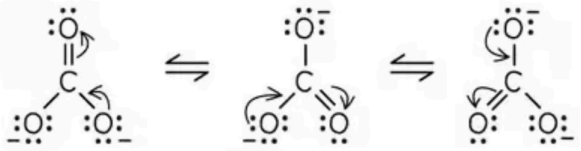
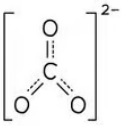
- One the resonance structures are identified, the resonance hybrid can be drawn using **dotted lines** to show the position of delocalised electrons
- The three resonance structures have more energy than the resonance hybrid of nitrate(V) ion
  - Therefore, the resonance hybrid is is the most stable and the actual structure of the nitrate(V) ion

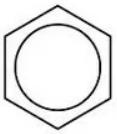
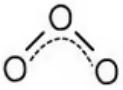
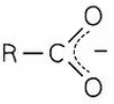
The Nitrate(V) Ion



Resonance hybrid nitrate(V) ion

- Some common examples of resonance are the carbonate ion, benzene, ozone, and the carboxylate ion which are showed in the table below

Species	Lewis Diagram	Resonance Hybrid
Carbonate ion, CO_3^{2-}		

Benzene, C_6H_6		
Ozone, O_3		
Carboxylate ion, $RCOO^-$		

Your notes

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The Octet Rule & Formal Charge

- Atoms form chemical bonds to get the electron configuration of their closest noble gas
- The **octet rule** explains why atoms usually want **eight electrons** in their valence shell in order to have more stability
- In some cases, it is possible to have multiple Lewis structures while still following the octet rule
- Each of these variations is a **resonance structure** and they contribute to the formation of the **hybrid of resonance**
- In order to figure out the resonance structure that contributes the most, the **formal charge** (FC) of all the atoms in the molecule is determined

Formal Charge

- **Formal charge** is a charge assigned to all the atoms in a molecule by assuming that all the bonding electrons are shared equally between the atoms, regardless of differences in **electronegativity**
- The formula of FC is

$$FC = V - \frac{1}{2}B - N$$

- *V* is the number of valence electrons
- *B* is the number of bonding electrons
- *N* is the number of nonbonding electrons

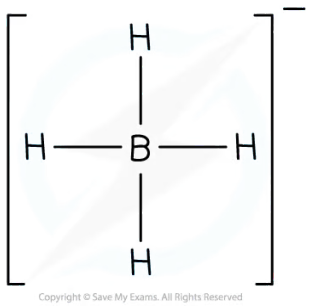
Worked Example

What is the formal charge of boron in the BH_4^- ion?

Answer:

- Boron is a Group 13 element, by using the periodic table information it can be deduced that it has 3 valence electrons.
- Hydrogen is in Group 1, therefore it has one valence electron.
- Since the BH_4^- ion has lost one electron, the total number of valence electrons in the structure is 8.
- By following the steps to draw a Lewis diagram, the structure should look like this:

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Lewis formula of BH_4^-

- The number of bonding electrons in B is 8 and the number of nonbonding electrons in B is 0

$$FC = V - \frac{1}{2}B - N$$

$$FC = 3 - \frac{1}{2} \times 8 - 0$$

$$FC = -1$$

Choosing the preferred resonance structure

- The concept of formal charge can help select competing resonance structures
- The following rules should be applied when analyzing **which of the resonance structures contributes the most**
  - The sum of the formal in neutral molecules is zero
  - The sum of the formal charges in ions is the charge of the ion
  - The smaller the formal charge (or 0), the better it is. Large formal charges are not common in atoms
  - Negative formal charges should be assigned to the most electronegative atom

Worked Example

What is the formal charge on the two resonance structures shown?



Resonance structures of hydrogen cyanide

Your notes

Deduce which is the preferred structure

Answer:

Structure 1

$$FC = V - \frac{1}{2}B - N$$

$$FC \text{ on hydrogen} = 1 - \frac{1}{2}(2) - 0 = 0$$

$$FC \text{ on carbon} = 4 - \frac{1}{2}(8) - 0 = 0$$

$$FC \text{ on nitrogen} = 5 - \frac{1}{2}(6) - 2 = 0$$

Structure 2

$$FC = V - \frac{1}{2}B - N$$

$$FC \text{ on hydrogen} = 1 - \frac{1}{2}(2) - 0 = 0$$

$$FC \text{ on carbon} = 4 - \frac{1}{2}(6) - 2 = -1$$

$$FC \text{ on nitrogen} = 5 - \frac{1}{2}(8) - 0 = +1$$

- **Structure 1** is the preferred structure because
  - The sum of the formal charges is zero since its a neutral molecule
  - All the formal charges on the atoms are 0
  - It has the **least electronegative atom (the carbon atom) in the central position**
- **Structure 2** has +1 on N and -1 on C
  - The negative formal charge in **Structure 2** is not on the most electronegative ion, therefore, the structure is unstable

Examiner Tips and Tricks

The terms Lewis Diagrams, Lewis Structures or Lewis Formulas mean the same thing

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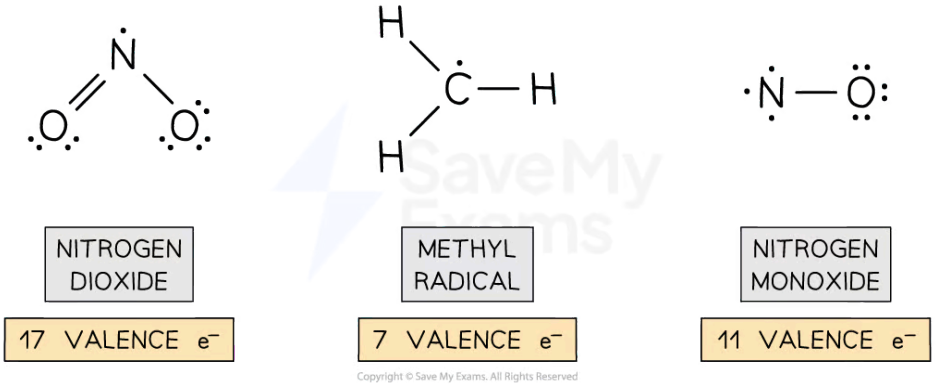
Limitations of Lewis Structures

- There are some chemical compounds that cannot be represented with Lewis diagrams accurately
- This occurs because the elements involved do not follow the octet rule

Molecules with odd number of electrons

- One of the most common cases occurs when there is an **odd number of electrons** in the molecule
- There are actually few stable molecules in nature with odd number of electrons, because an molecules with **unpaired electrons are highly reactive and unstable**
- These odd electron species are called **free radicals**

Examples of Free Radicals



Molecules with an odd number of electrons

Molecules with expanded valence shells

- There are some molecules that exceed the eight electrons in their valence shell
- An atom with an **expanded octet** can hold up to ten electrons, or even twelve
- These expansion occurs because there are empty f-orbitals available for bonding after the third energy shell
  - Therefore, the third period elements can occasionally exceed the octet to accommodate additional electrons
- Expanded octets occurs occasionally when a **non metal central atom from the third period or below**, is surrounded by **terminal atoms with high electronegativity values**
- The most common atoms that form expanded octets are: S and P

The PCl_5 molecule

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The octet of the central phosphorus atom has been expanded to hold 10 electrons

The SF_6 molecule



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The octet of the central sulfur atom has been expanded to hold 12 electrons

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